



DIPARTIMENTO DI ECCELLENZA
MUR 2023/2027
A.D. 1308
unipg
DIPARTIMENTO
DI FISICA E GEOLOGIA



ASTRI Data challenge

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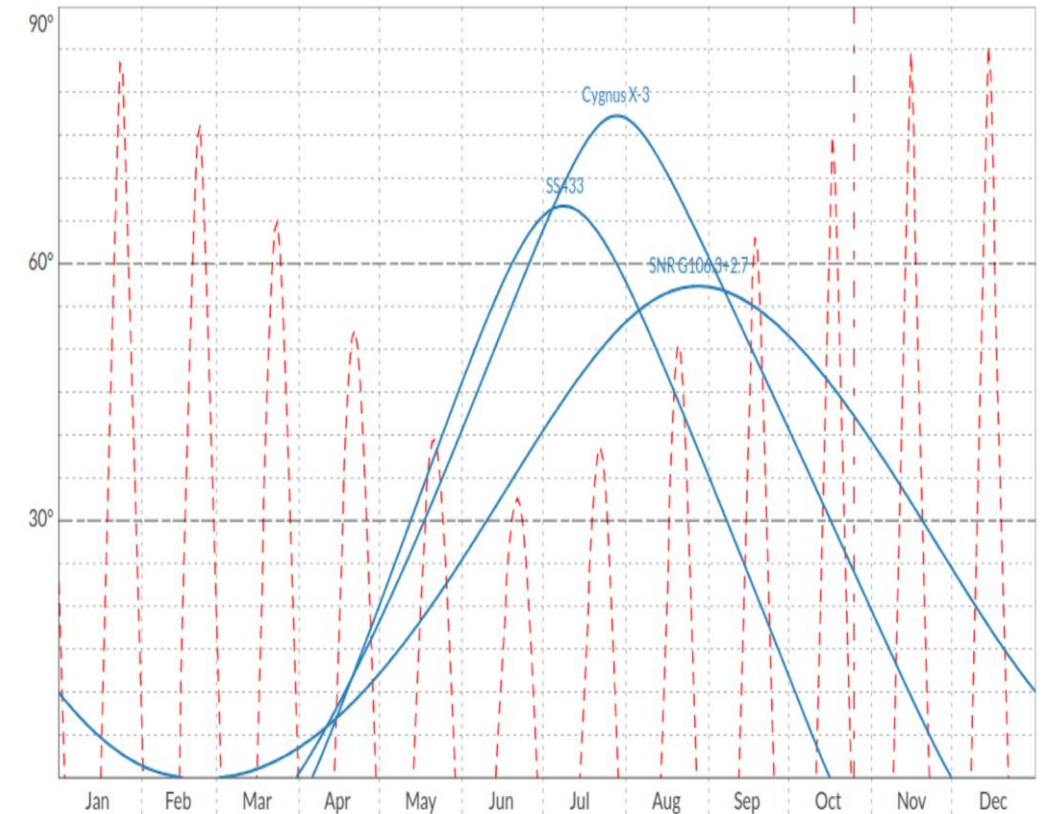
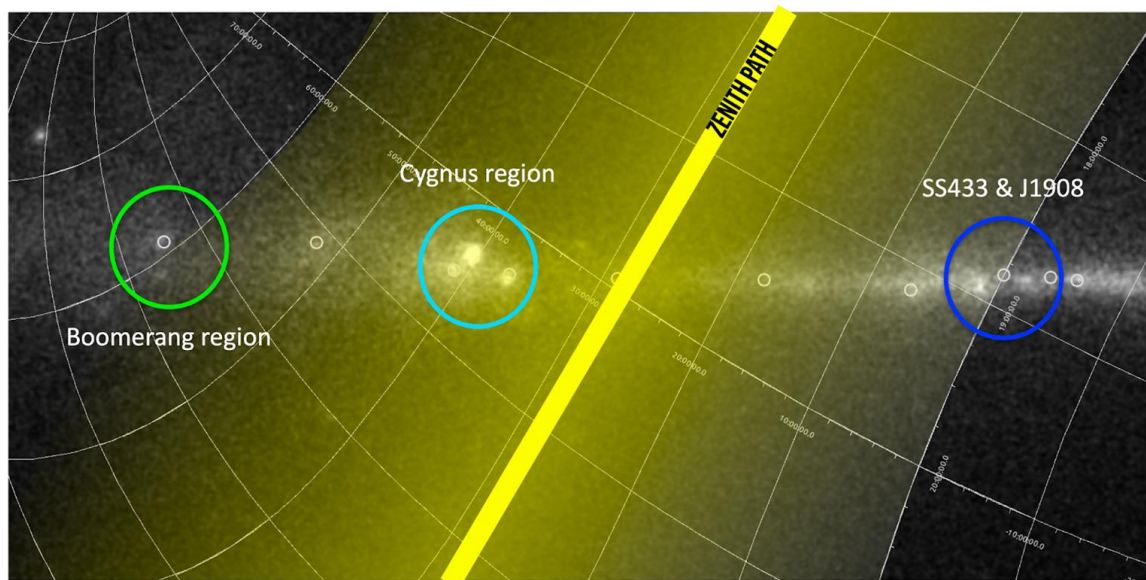
23/04/2026

WHAT WE WANT TO DO

- We want to be ready for the Cygnus observations
- Realistic simulations of the Cygnus region are required:
 - **First observations in stereoscopic mode**
 - **pointing strategy**
 - staring pointing,
 - wobble pointing,
 - scanner
 - **understanding sources** visible with ASTRI

Cygnus Visibility

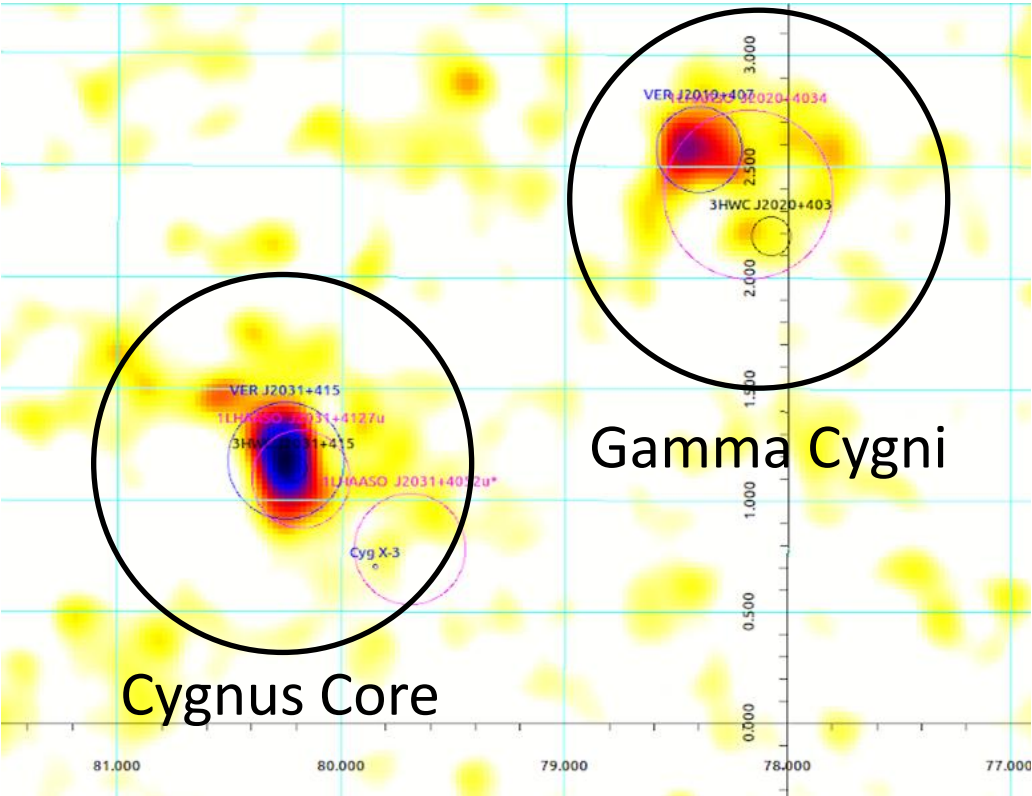
- Cygnus will be perfectly visible from June to October 2026
- 400 hours available



SIMULATIONS

V0.1:

- ASTRI IRF 3T
- 401 runs
- $t_{\text{obs}} = 1 \text{ h}$
- Wobble mode:
 - Pnt1: $(79.57^\circ, 2.16^\circ)$
 - Pnt2: $(78.86^\circ, 1.45^\circ)$
- BKG : randomised
- Energy range: (1, 250) TeV



	Zenith angle	Jun	Jul	Aug	Sep	Ott	TOT
Cygnus	0–20 °	66h	96 h	99 h	86 h	54h	401 h

Cygnus Region

LHAASO Cygnus Paper

An Ultrahigh-energy γ -ray Bubble Powered by a Super PeVatron

LHAASO Collaboration^{1,2}

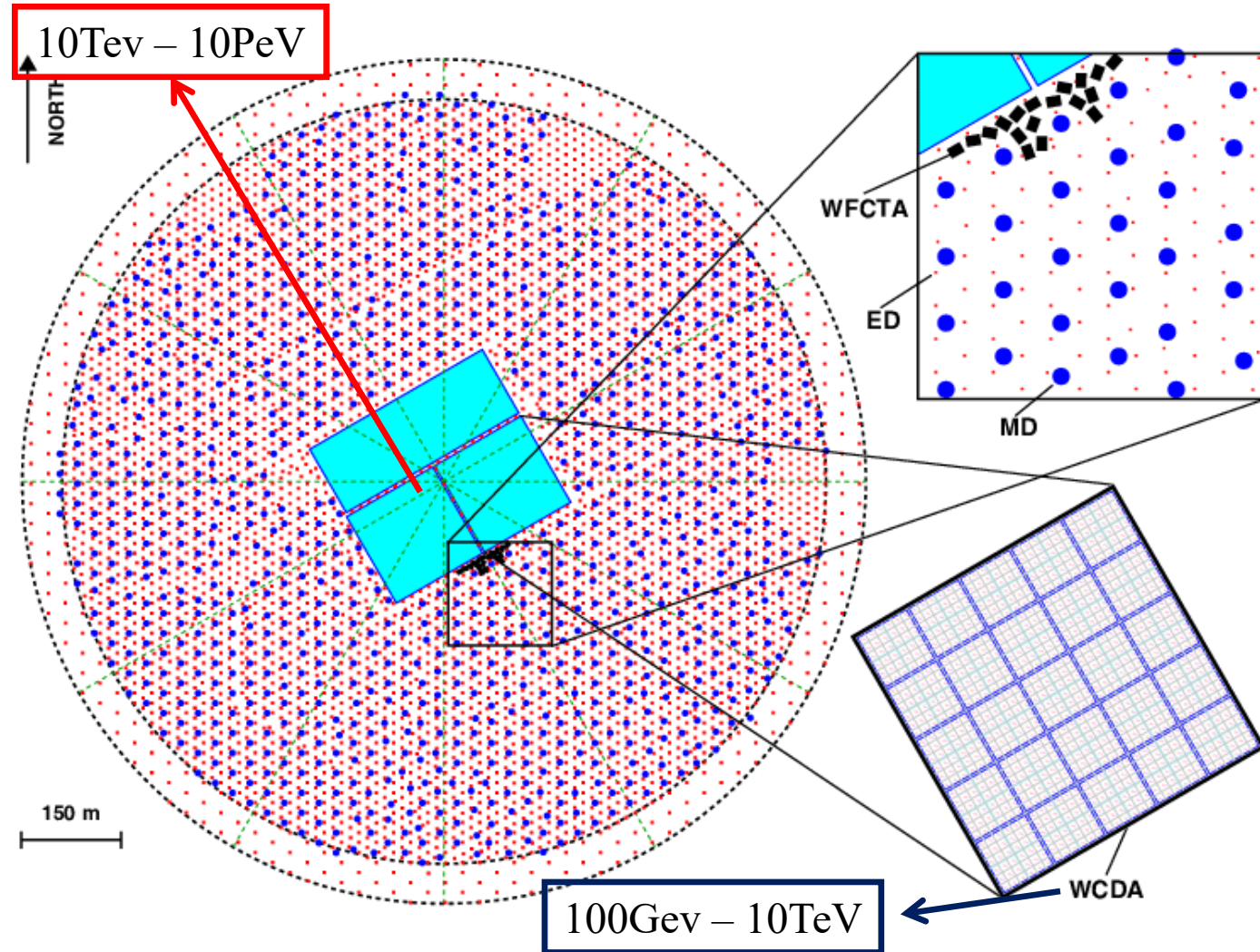
Abstract

We report the detection of a γ -ray *bubble* spanning at least 100 deg^2 in ultra high energy (UHE) up to a few PeV in the direction of the star-forming region Cygnus X, implying the presence Super PeVatron(s) accelerating protons to at least 10 PeV. A log-parabola form with the photon index $\Gamma(E) = (2.71 \pm 0.02) + (0.11 \pm 0.02) \times \log_{10}(E/10 \text{ TeV})$ is found fitting the gamma-ray energy spectrum of the bubble well. UHE sources, hot spots correlated with very massive molecular clouds, and a quasi-spherical amorphous γ -ray emitter with a sharp central brightening are observed in the bubble. In the core of $\sim 0.5^\circ$, spatially associating with a region containing massive OB association (Cygnus OB2) and a microquasar (Cygnus X-3), as well as previously reported multi-TeV sources, an enhanced concentration of UHE γ -rays are observed with 2 photons at energies above 1 PeV. The general feature of the bubble, the morphology and the energy spectrum, are reasonably reproduced by the assumption of a particle accelerator in the core, continuously injecting protons into the ambient medium.

Cygnus Region

Introduction

Cygnus-X is one of the most intensive and nearby (at a distance ≈ 1.4 kpc) star-forming regions in the Milky Way. The $7^\circ \times 7^\circ$ area harbors several Wolf-Rayet stars and hundreds of O-type stars grouped in powerful OB associations. It also contains vast HI and molecular gas complexes with masses exceeding $10^6 M_\odot$. The presence of potential particle accelerators, namely massive stars and supernova remnants, and targets for γ -ray production (dense gas regions) make several parts of this region effective γ -ray emitters. Fermi-LAT has detected high energy γ -rays from the direction of the massive star association Cygnus OB2. The extended $\sim 2^\circ$ source (dubbed ‘Cygnus Cocoon’ [1]) later has been reported at TeV [2, 3] and PeV photo was also detected from this direction [4] energies. The latter has attracted particular attention due to its potential connection with the origin of the highest energy galactic CRs around and beyond the so-called ‘knee’. Here, we present the results of new observations of the Cygnus region with the Large High Altitude Air Shower Observatory (LHAASO) and discuss their astrophysical implications.



Cygnus Region

Observation

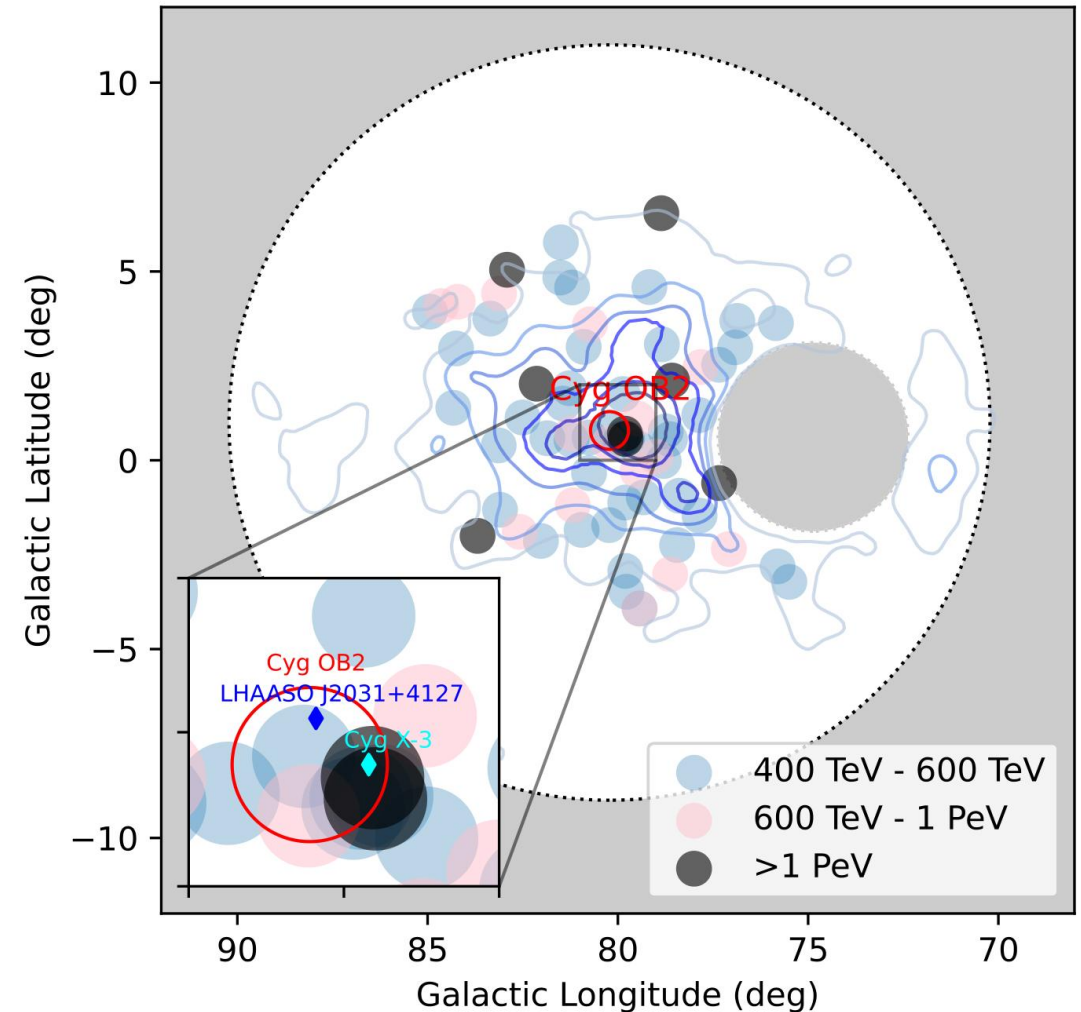
2. The LHAASO Observation

Our new observations firstly confirm the existence of a very extended emission at Cygnus region in ultra-high energy band. The discovery beyond our initial publication[4] includes a few striking findings, namely (i) the detection of an enormous γ -ray bubble that extends to more than 6° from the center, (ii) the detection of hot spots associated with massive molecular clouds, (iii) the observation of the SED of the bubble up to 2 PeV, and (iv) finding a new source nearly at the center of the bubble, spatially coincident with both Cygnus OB2 and Cyg X-3.

Cygnus Region

Events and sources

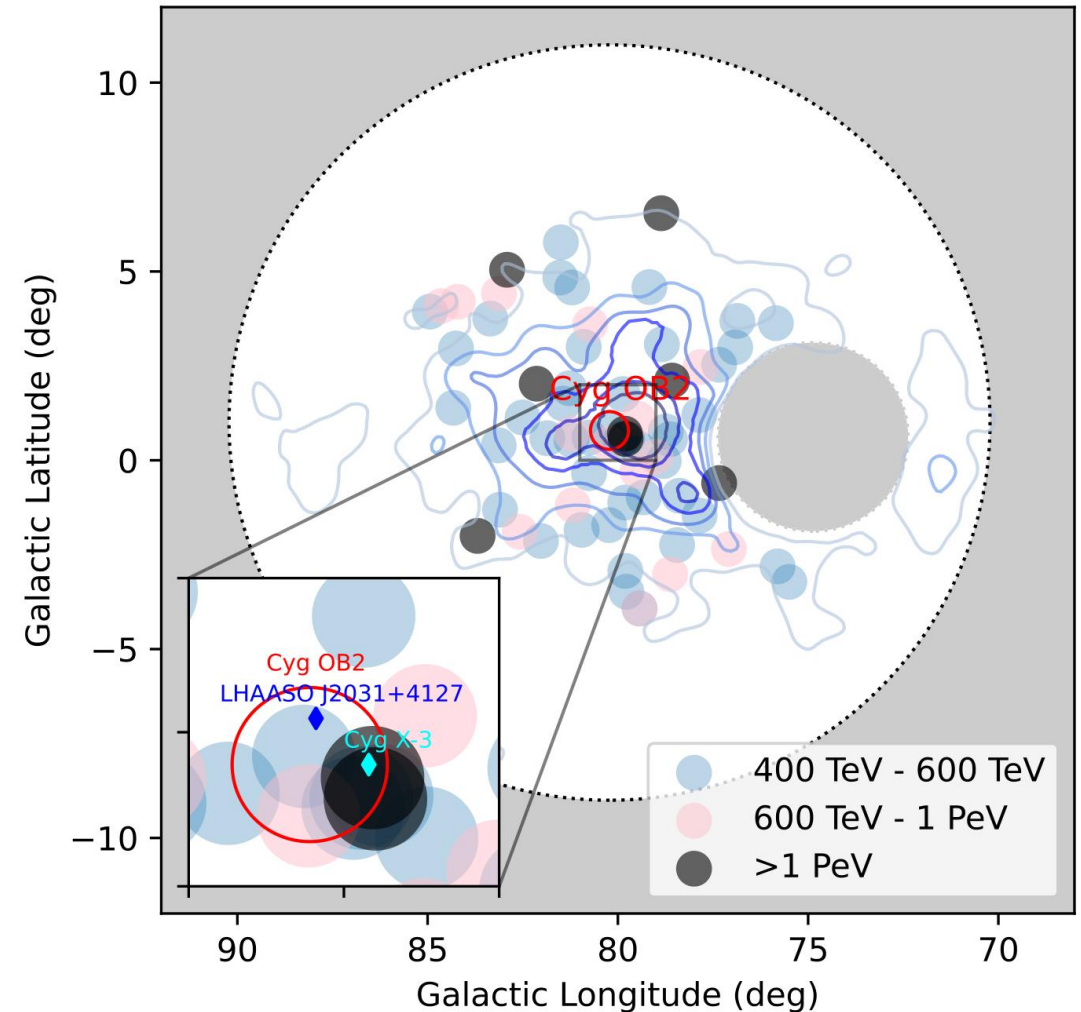
Figure 1 shows the contour map of 2 to 20 TeV *gamma*-rays obtained towards Cygnus X, and 66 individual photon-like events of energies exceeding 400 TeV. The estimated cosmic ray background is about 9.5, and the contamination will be smaller with the increase of energy benefiting from the improvement of rejection power. There are 8 events with energy above 1 PeV, whileas the background is only 0.75. Considering the small background-to-signal ratio, we get valuable information from the individual events. They are distributed inhomogeneously. In general, the density is higher at a closer distance to the core. Especially within 0.5° radius around Cygnus OB2, 7 events above 400 TeV have been detected, including two photons of energies exceeding 1 PeV. This implies the operation of central CR accelerator(s) that injects relativistic protons and nuclei into the circumstellar medium (see Supplementary Section).



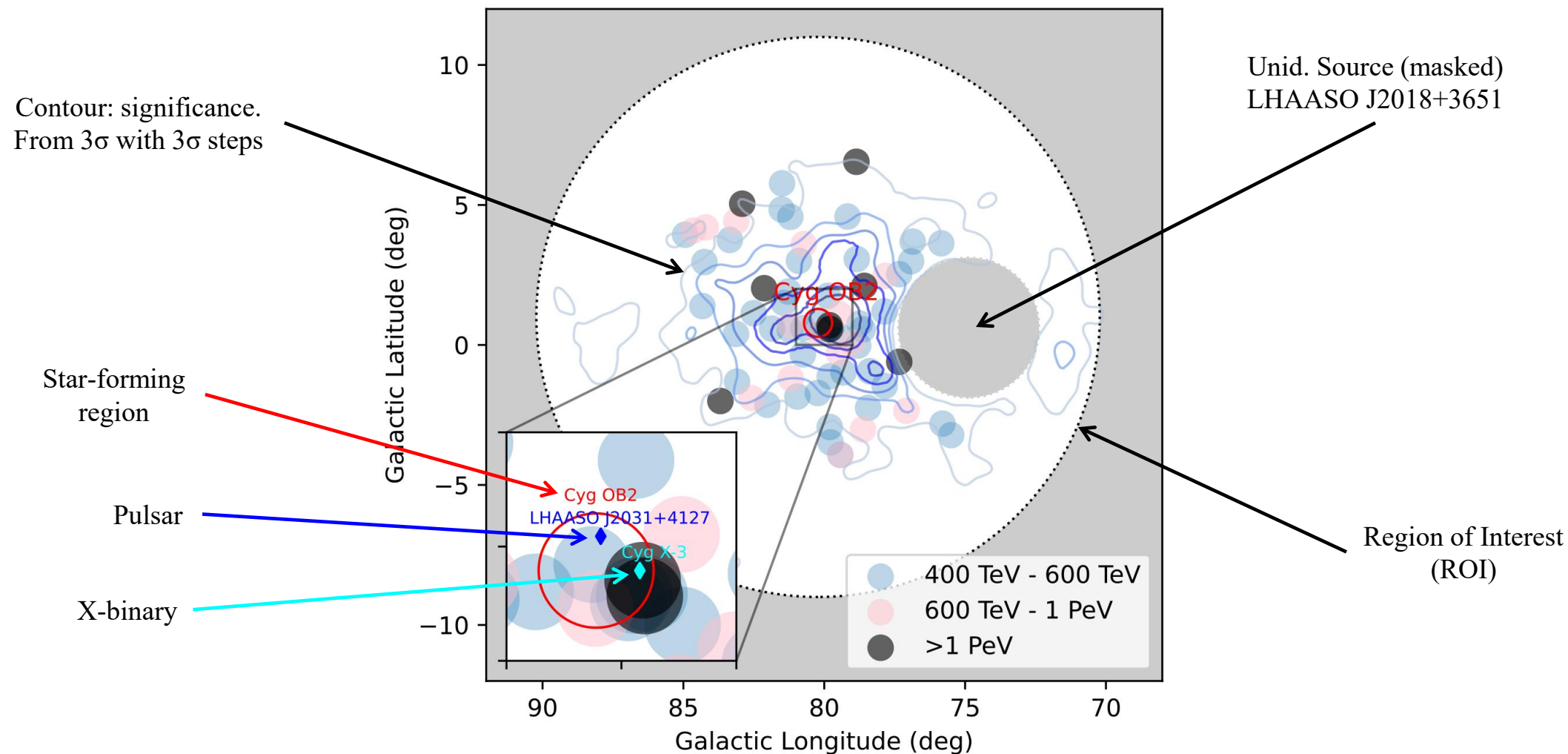
Cygnus Region

In this relatively compact central region with a radius of 0.5° (hereafter, the core) are located Cygnus OB2, a massive OB association, and the powerful X-ray binary Cyg X-3. There are three TeV sources, including the the hard-spectrum TeV source TeV J2032+4130 [7] (=LHAASO J2031+4127), binary system PSR J2032+4127/MT91 213[8] and the ultra-high energy source LHAASO J2032+4102[4], are detected in this region. The resolved sources potentially correlated with previous TeV sources are discussed in supplementary material. A new source (LHAASO J2031+4057), detected by WCDA below 10 TeV, is also found in the core region, in spatial association with both Cygnus OB2 and Cyg X-3. The similarity of spectral shape with the extended emission suggests that this source is presumably a part of the bubble. The increase of TS by adding a new source at the core region is 18 for KM2A, which shows a hint of excess but it is not significant now.

Immediately outside the core is a TeV γ -ray source, which is associated with the middle-aged supernova remnant γ -Cygni. The detailed study of this interesting object, which overlaps with the bubble, will be published elsewhere. In this paper,



Cygnus Region

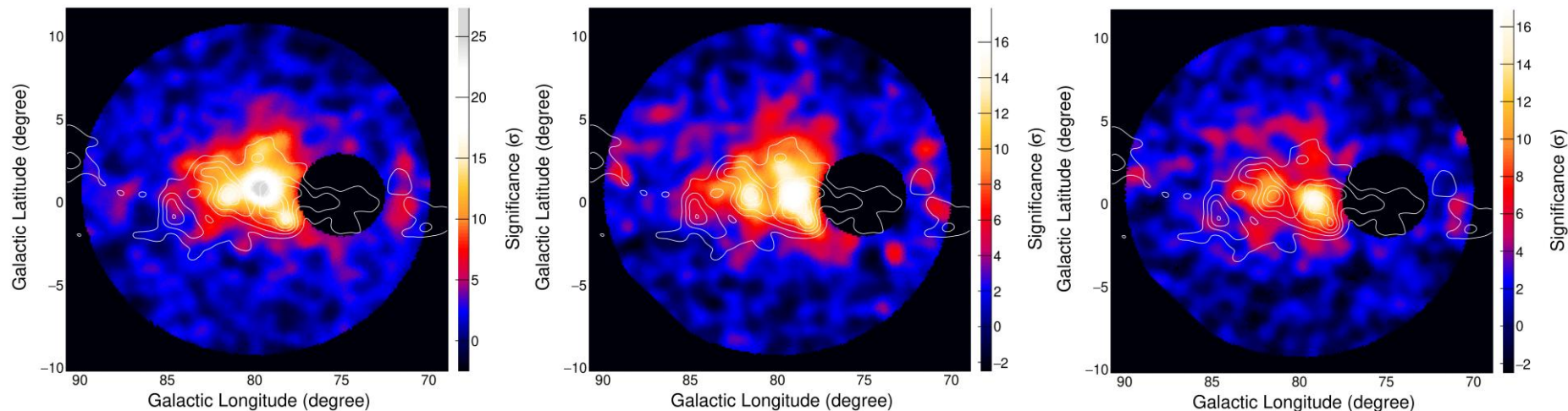


Cygnus Region

The Cygnus Bubble

After the removal of all (identified and unidentified) γ -ray sources and ‘hot spots’, a giant γ -ray structure (hereafter, the Cygnus Bubble) reveals both in the WCDA and KM2A data. The residual structure, detected from ~ 2 TeV to ≥ 1 PeV and spreading to $\sim 10^\circ$, formally is fitted with a Gaussian template for the inner part together with an extended tail approximately proportional to the HI angular distribution pattern. The

In principle, the angular distribution of the bubble emission could be a result of the superposition of a very large number of unresolved discrete *gamma*-ray sources, but such a concentration of UHE γ -ray emitters in a relatively small volume hardly could be the case. More realistically, it seems to be linked to the local cosmic rays, i.e., relativistic protons and nuclei accelerated inside the bubble and distributed with enhanced density in the inner parts of the bubble.

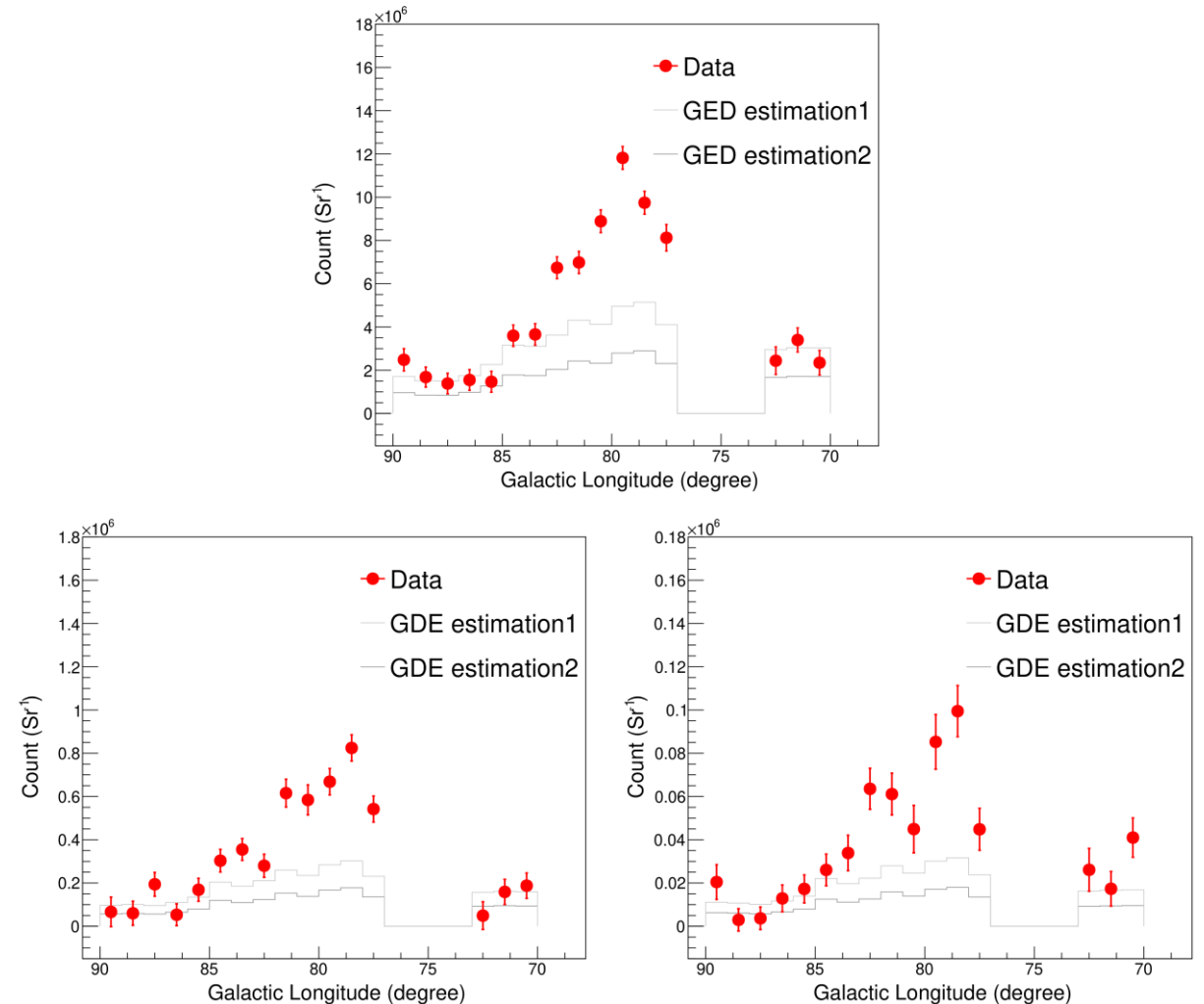


NB: here contour is the local molecular cloud distribution

Cygnus Region

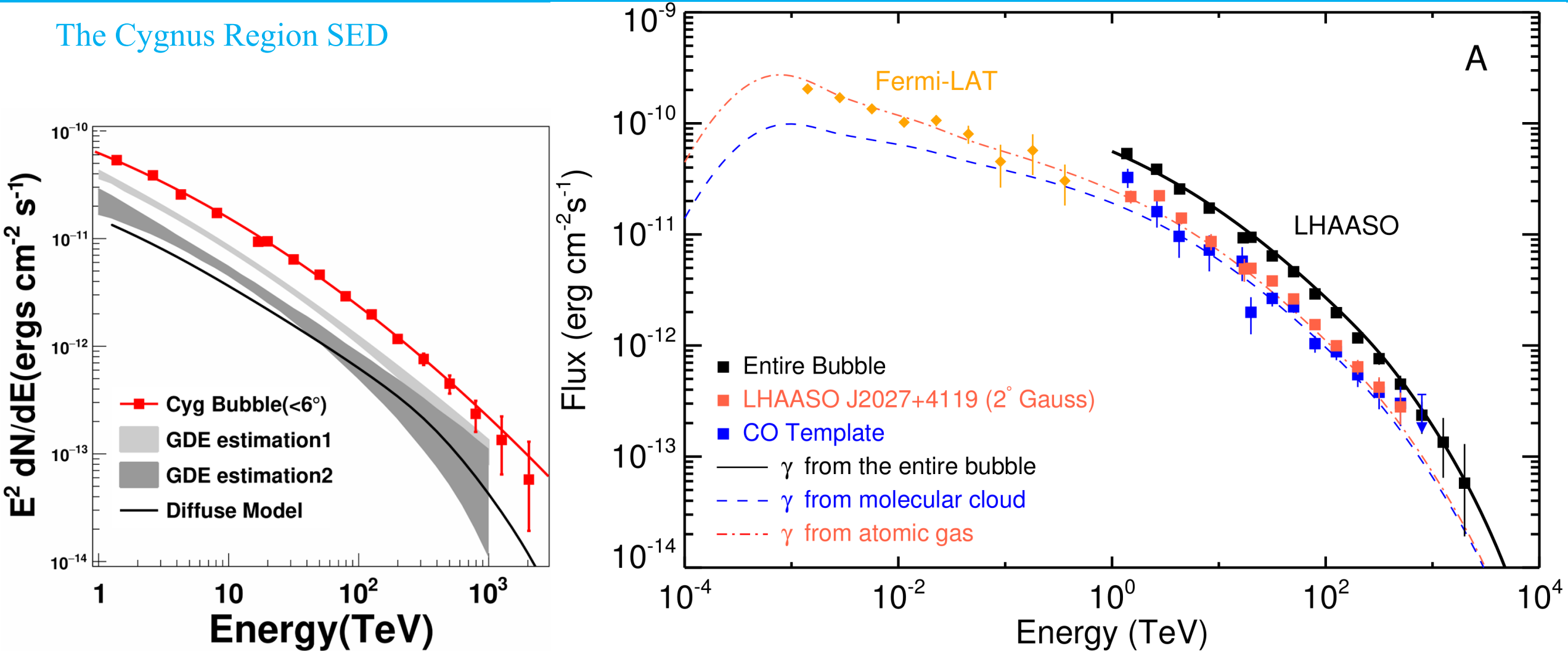
The Cygnus Bubble

The distribution of the excess counts and the estimated Galactic diffuse γ -ray emission (GDE) as functions of the galactic longitude are shown in Figure 2 (the bottom row of panels). It is clear that the γ -ray brightness distribution is much sharper than the distribution of GDE. This excludes a significant contribution of GDE to the flux coming from the bubble itself. Indeed, the GDE is produced in interactions of the cosmic ray ‘sea’ with the interstellar gas throughout the Galaxy, which is proportional to the total gas column density in the given direction, $N_{\text{HI}+\text{H}_2}(\theta)$. The measured γ -ray fluxes substantially exceed, by a factor of 2-3, any realistic estimate of the GDE flux (see below). The bubble-to-GDE ratio clearly increases towards the center of the bubble.



Cygnus Region

The Cygnus Region SED



Mini Data Challenge

Goal: implement a blind search approach to

1. identify the source models and their associated spectral properties
2. reconstruct the background model

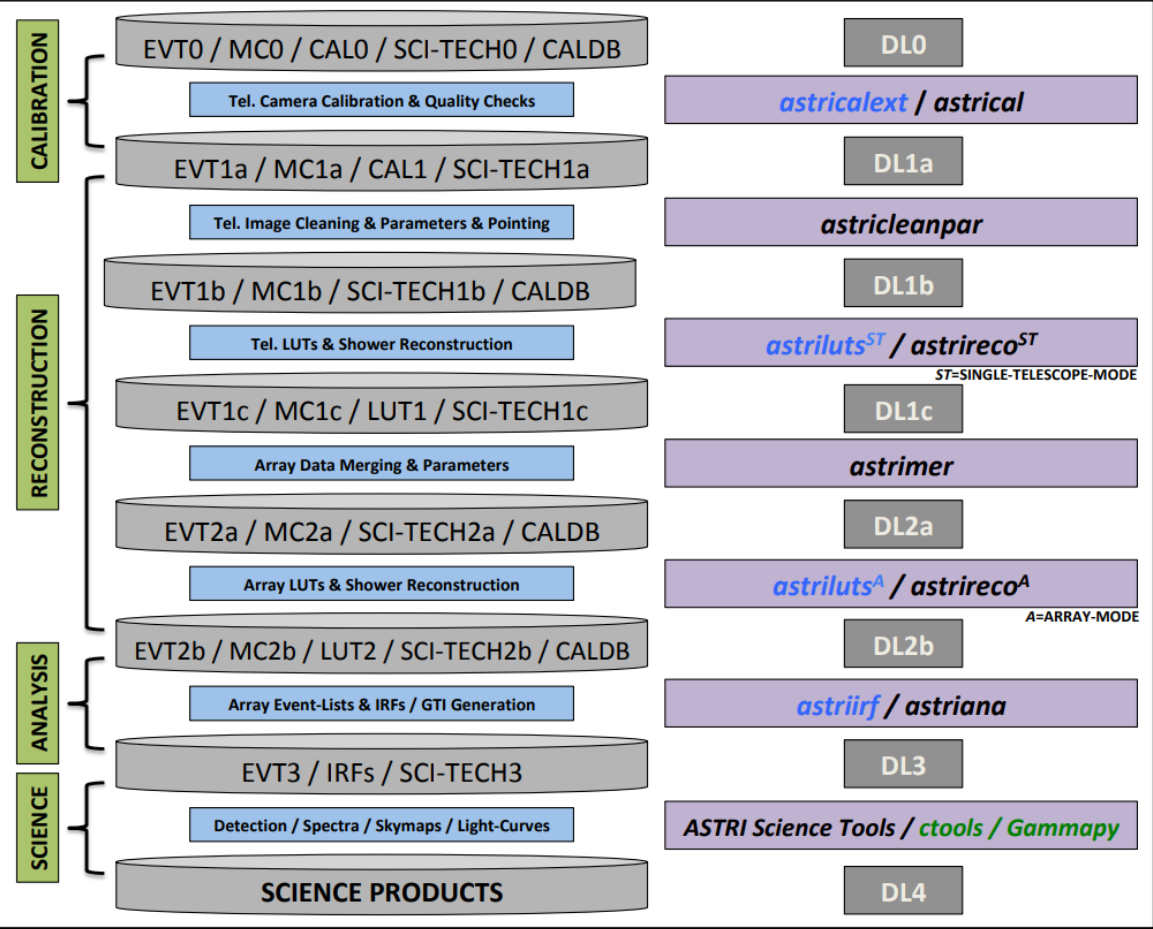
What we have: Simulated data (version V0.1) and IRF

1. Events folder: 401.fits files with EVENTS data
2. MA_102_008_zd20-20_az000and180_0502_cc03_TV_v0.1.lv3.fits: IRF
3. Pointings_wobble.fits: pointing info

Mini Data Challenge

How the file system works in ASTRI

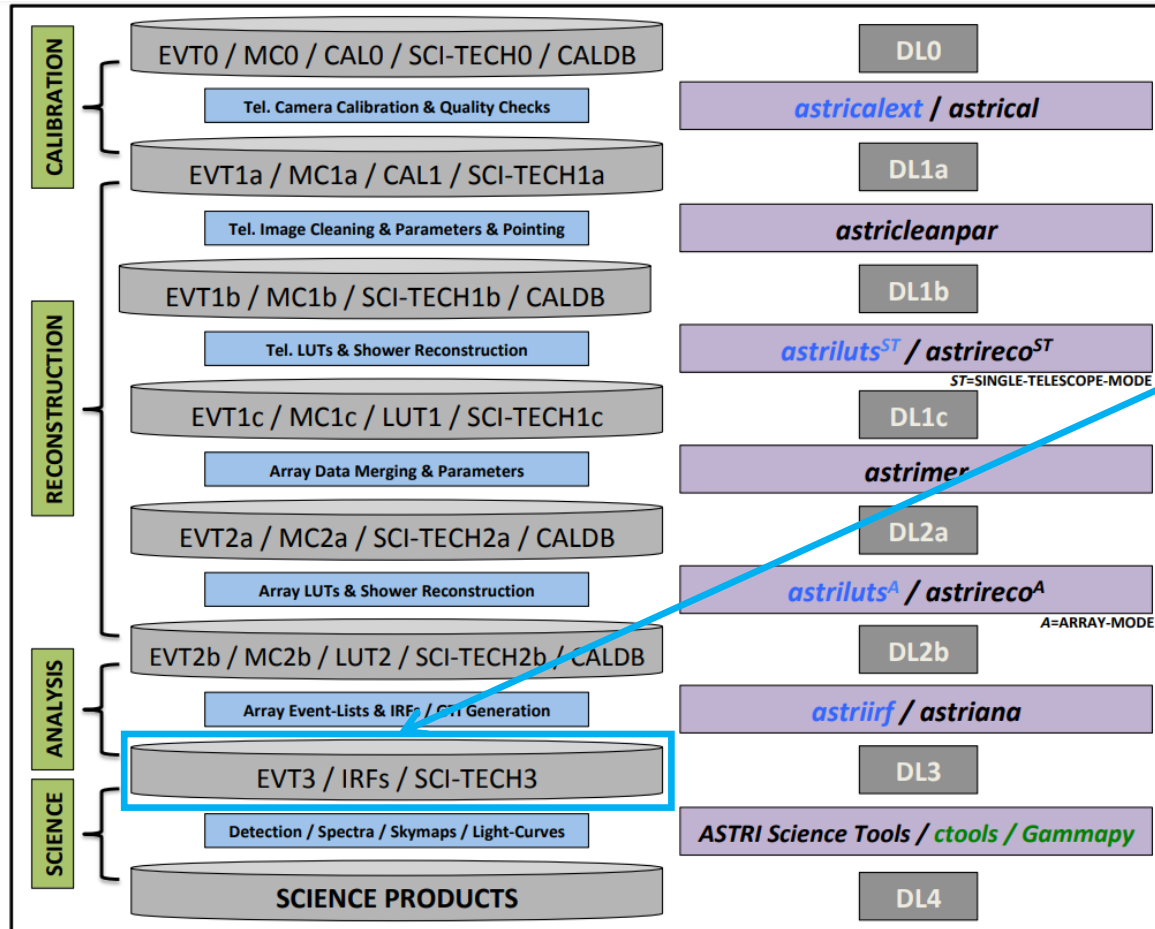
<https://openaccess.inaf.it/server/api/core/bitstreams/17a9e90e-66e0-4860-8e3a-4dbf213227ec/content>



Mini Data Challenge

How the file system works in ASTRI

<https://openaccess.inaf.it/server/api/core/bitstreams/17a9e90e-66e0-4860-8e3a-4dbf213227ec/content>



Here we are!

- **Analysis (DL2b → DL3):**

The fully reconstructed array-wise data (EVT2b/MC2b) are further processed by the *astriirf* and *astriana* modules to achieve the fully reduced data (EVT3/IRF3). The former module is aimed at computing the global instrumental response functions (IRF2), which include the effective collection area, energy and angular resolution, and residual background rate (as a function mainly of energy, zenith and azimuth pointing, and gamma-ray source position). The latter module is then in charge of providing the final event-list (EVT3), extracted from the corresponding EVT2b, along with the corresponding reduced IRF3 (from the corresponding IRF2). In this last analysis step, both quality and gamma/hadron separation cuts are applied to get the final DL3 data.

- **Science (DL3 → DL4):**

The fully reduced data (EVT3/IRF3) are finally analyzed with either science tools currently being used in the CTA Consortium (e.g. *ctools*³³ and *Gammapy*³⁴) or with specifically developed ones (*ASTRI Science Tools*). The final scientific products (DL4), such as detection plots, spectra, sky-maps, and light-curves, are thus generated. These products, if needed, can be further processed and merged to get high-level observatory data (DL5), which include legacy observatory data, such as survey sky-maps and/or source catalogs.

Mini Data Challenge

Events files (0201_ccXX.lv3):

1. cc03: 50h, cuts maximizing sensitivity
2. 401 runs, then 401 .fits files, one per acquisition hour
3. each file has 2 tables EVENTS and GTI:
 1. EVENTS:
 - ENERGY_TRUE (TeV) and ENERGY (TeV)
 - RA_TRUE (deg) and RA (deg)
 - DEC_TRUE (deg) and DEC (deg)
 - MC_ID
 - EVENT_ID
 - DETX (deg)
 - DETY (deg)
 2. GTI are the Good Time Intervals (run-based) data produced by Data Quality Check pipeline and associated to EVT3 by ASTRIANA
 - START and STOP (s)

Mini Data Challenge

Pointing Wobble files:

1. 2 pointings: 200 runs with pnt1 (79.57°, 2.16°), 201 runs with pnt2: (78.86°, 1.45°)
2. POINTINGS_WOBBLE:
 - Tsec (s)
 - MJD (d)
 - GLON (deg)
 - GLAT (deg)
 - nsb (mag/arcsec²)
 - source_alt (deg)
 - moon_alt (deg)
 - moon_phase (deg)
 - moon_sep (deg)

Mini Data Challenge

IRFs: Effective Area, Point Spread Function, Energy Dispersion, Background

- Energy range: (1, 250) TeV
from Data Challenge presentation

● LV3 selection (lv2b → lv3):

- 16 reconstructed energy bins between [0.2-316] TeV (5 energy bins per decade)
- 300 true energy bins between [0.01-10000] TeV (50 energy bins per decade)
- 1 bin of AZ/ZD ($ZD < 30^\circ$)
- 6 bins of offset between $[0^\circ, 6^\circ]$
- class cuts (gammaness) based on (gamma) efficiency: 95%, 90%, 80%, 70%, 60%, 50% (aka: cc10, cc11, cc12, cc13, cc14, cc15) (cc11 and cc14 default class cuts)

from Saverio Lombardi's hands on presentation

lv3 standard selection
for pass_v0.0.2

Mini Data Challenge

IRFs: **Effective Area**, **Point Spread Function**, **Energy Dispersion**, **Background**

- ENERGY_HI (TeV), shape = (16)
- ENERGY_LO (TeV), shape = (16)
- THETA_HI (deg), shape = (6)
- THETA_LO (deg), shape = (6)
- EFFAREA (m²), shape = (16, 6)
- THETA_HI (deg), shape = (6)
- THETA_LO (deg), shape = (6)
- MIGRA_HI, shape = (300)
- MIGRA_LO, shape = (300)
- MATRIX, shape = (300, 300, 6)
- THETA_HI (deg), shape = (6)
- THETA_LO (deg), shape = (6)
- SCALE (1/sr), shape = (16, 6)
- SIGMA_1 (deg), shape = (16, 6)
- SIGMA_2 (deg), shape = (16, 6)
- SIGMA_3 (deg), shape = (16, 6)
- AMPL_2, shape = (16, 6)
- AMPL_3, shape = (16, 6)
- DETX_HI (deg), shape = (60)
- DETX_LO (deg), shape = (60)
- DETY_HI (deg), shape = (60)
- DETY_LO (deg), shape = (60)
- BKG (1/s 1/MeV 1/sr), shape = (16, 60, 60)